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CONCEPT MAP

Oscillation

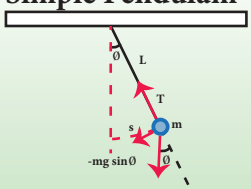
Simple Harmonic Motion (SHM)

$F = -kx \quad x = +A \sin\omega t$

$\frac{d^2x}{dt^2} + \omega^2 x = 0$

- $a = -\omega^2 A \sin\omega t$, $a = -\omega^2 y$
- $V = A\omega \cos\omega t$
  - $v = \omega\sqrt{A^2 - y^2}$
- $F = -\frac{dU}{dx}$ — $U = \frac{1}{2}kx^2$
- $TE = \frac{1}{2}kA^2$ — $KE = \frac{1}{2}mv^2$
- $T = \frac{2\pi}{\omega}$
  - Angular SHM: $\omega = \sqrt{\frac{\kappa}{I}}$
  - Linear SHM: $\omega = \sqrt{\frac{k}{m}}$

Simple Pendulam



Differential equation:

$$\frac{d^2\theta}{dt^2} = -\frac{g}{l}\theta$$

Time Period:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

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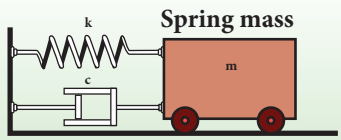
Differential equation:

$$\frac{d^2y}{dt^2} = -\frac{2g}{l}y$$

Time Period:

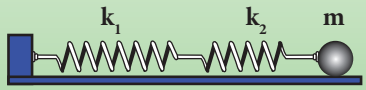
$$T = 2\pi\sqrt{\frac{l}{2g}}$$

Spring mass



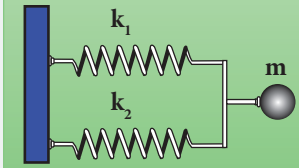
Combination of Springs

(1) Series:



$$k_s = \frac{k_1 k_2}{k_1 + k_2}$$

(2) Parallel:



$$k_p = k_1 + k_2$$